

A hand-drawn rocket model is shown on a chalkboard background. The rocket is constructed from various materials: a green body with orange bands, brown cardboard fins, and a cluster of colorful pencils (red, yellow, purple, green, black) forming the engine. A crumpled red and orange plastic bag is attached to the bottom, representing exhaust. The text 'EOC240.03' and 'IDENTIFY PARTS OF A ROCKET' is overlaid in white, with 'By:' below it.

# EOC240.03

## IDENTIFY PARTS OF A ROCKET

By:

# REMINDER

## CADET SUMMER TRAINING OPPORTUNITIES

Aviation Technology and Aerospace

Intro to Drill and Ceremonial

Intro to Fitness and Sports

Intro to Marksmanship

Intro to Military Band

Intro to Pipe Band

Intro to Survival

Each about 1-2 weeks

# Teaching Points

- TP1 Explain the Parts of a Rocket
- TP2 Name the Parts of a Rocket



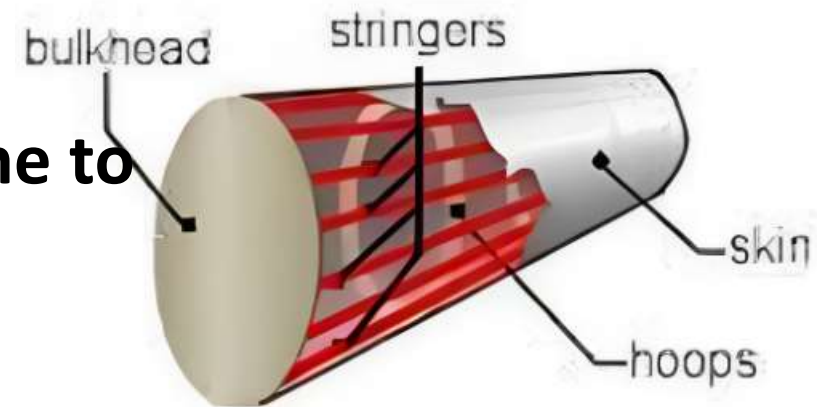
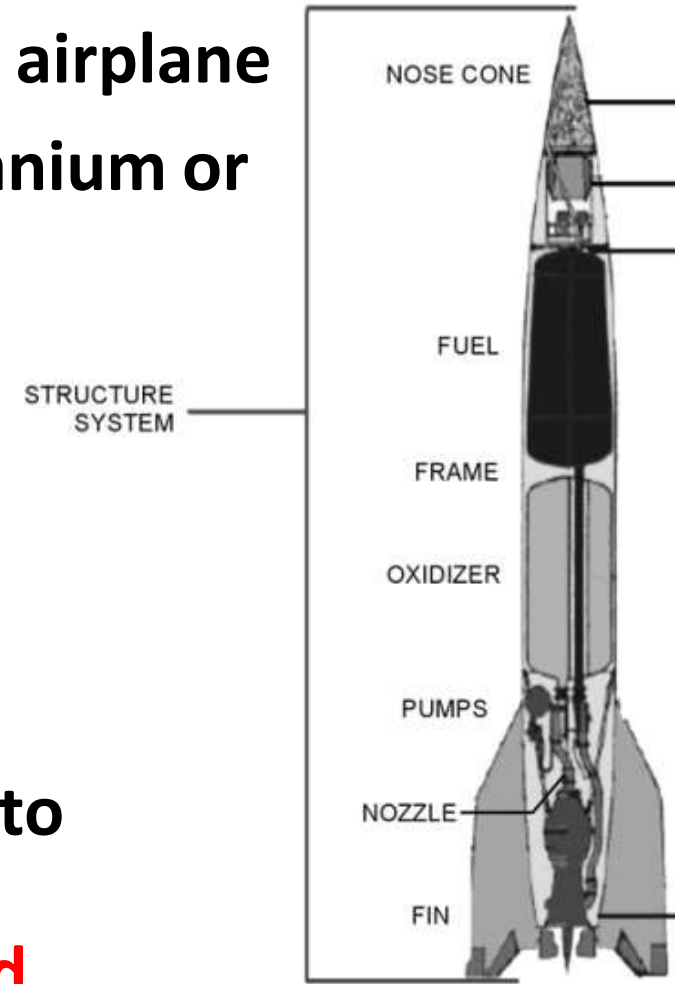


# TP1 Explain the Parts of a Rocket

- In flight, a rocket is subjected to the forces of **weight, thrust, and aerodynamics**
- There are many parts that make up a rocket.
- For design and analysis, engineers group parts that have the **same function into systems**



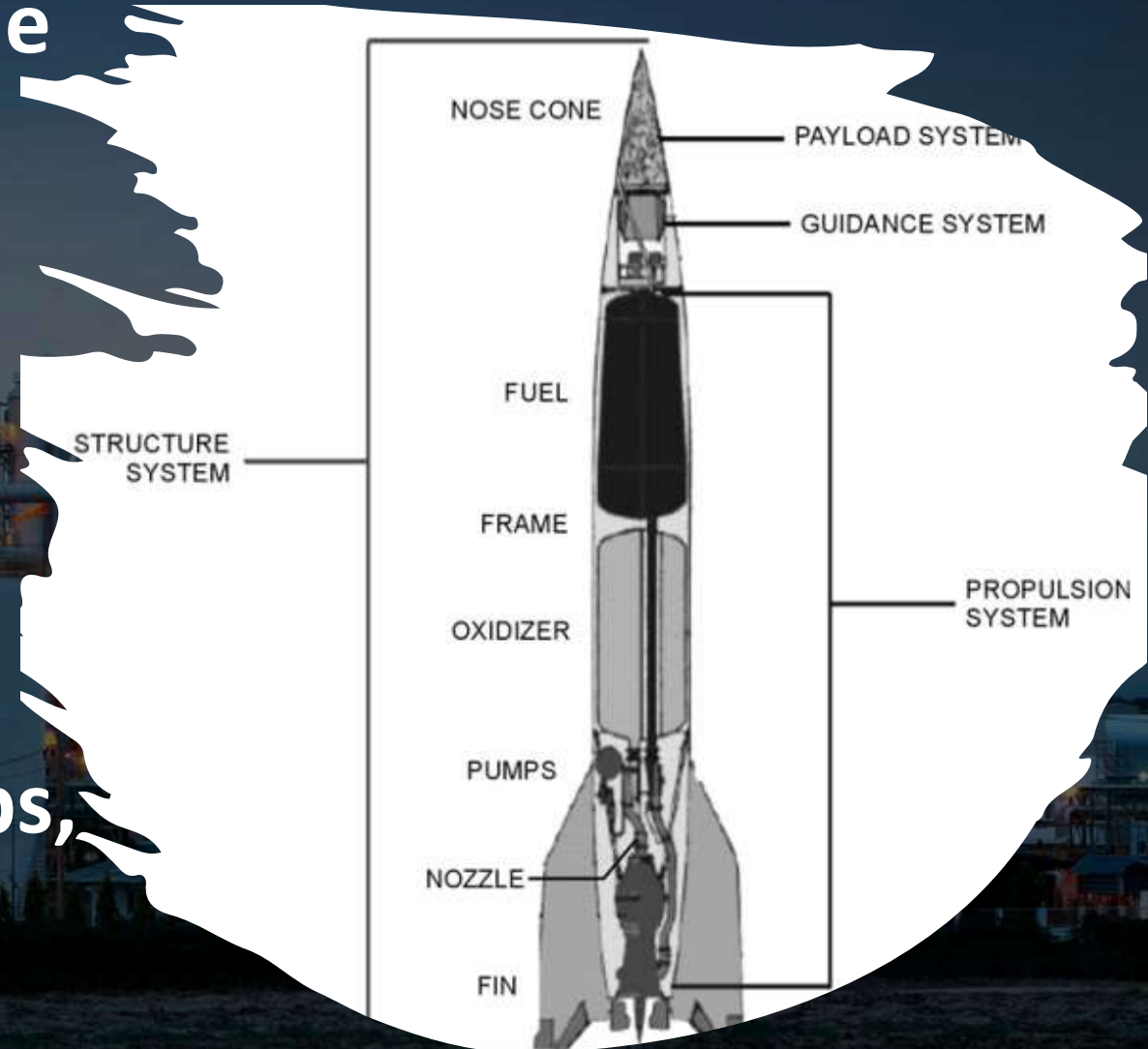
- Also known as the frame, it is similar to the fuselage of an airplane
- Made from **very strong but light-weight materials**, like titanium or aluminum
- The frame employs long **“stringers”** that run from the top to the bottom, which are connected to **“hoops”** that run around the circumference
- The **“skin”** is then attached to the stringers and hoops to form the basic shape of the rocket
- The skin may be coated with a thermal protection system to **keep out the heat of air friction** during flight and to **keep in the cold temperatures needed for certain fuels and oxidizers**
- Fins are attached to rockets at the bottom of the frame to **provide stability during flight**



## Structural System

# Propulsion System

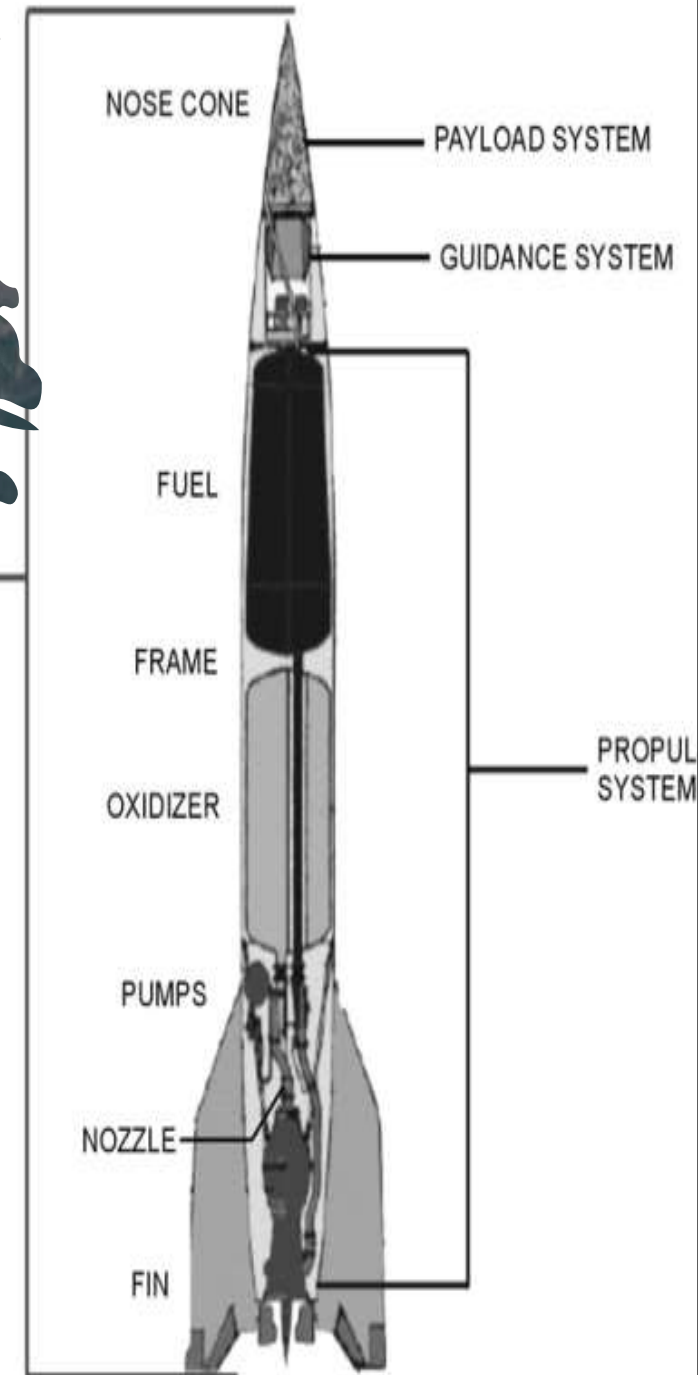
- Most of a full-scale rocket is made up of the propulsion system
- Two main propulsion systems
- **Liquid rocket engines** and **solid rocket engines**
- The V2 used a liquid rocket engine consisting of fuel and oxidizer (propellant) tanks, pumps, a combustion chamber with nozzle, and the associated plumbing.





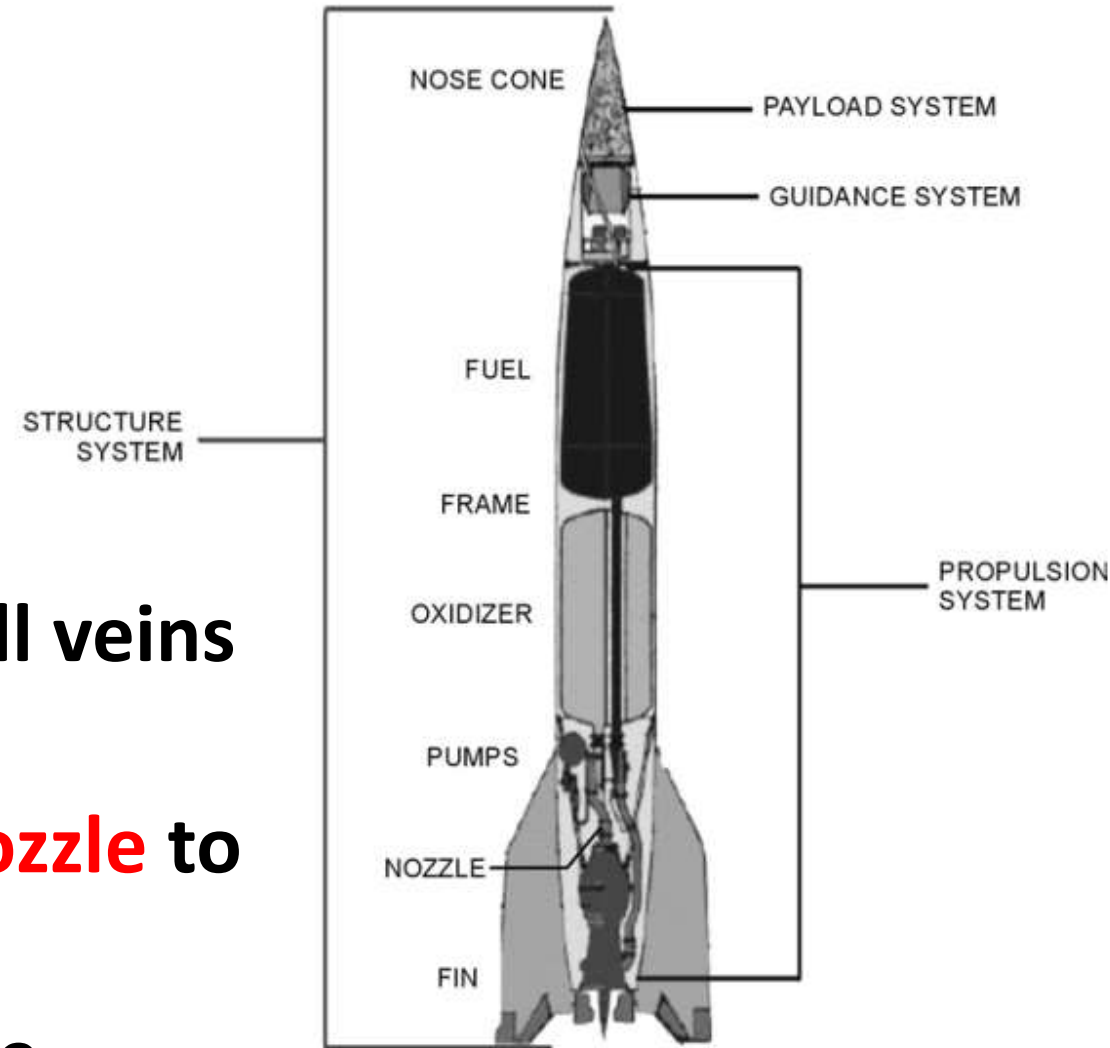
# Payload System

- Payload systems depend on the rocket's mission
- Following World War II, many countries developed guided ballistic missiles armed with nuclear warheads for payloads
- The same rockets were modified to launch satellites with a **wide range of missions**; communications, weather monitoring, spying, planetary exploration, and observatories, like the Hubble Space Telescope
- Special rockets were developed to launch people into Earth's orbit and onto the surface of the Moon



# Guidance System

- Very sophisticated sensors  
On-board computers  
Radars  
Communication equipment
- The V2 guidance system included small veins in the nozzle to deflect the thrust
- Modern rockets typically **rotate the nozzle** to maneuver the rocket.
- The guidance system must also provide **some level of stability** so that the rocket does not tumble in flight.







## QUESTIONS

- Q1. What rocket parts make up the structure system?

**The nose cone**

**Fuel**

**The frame**

**The oxidizer**

**Pumps**

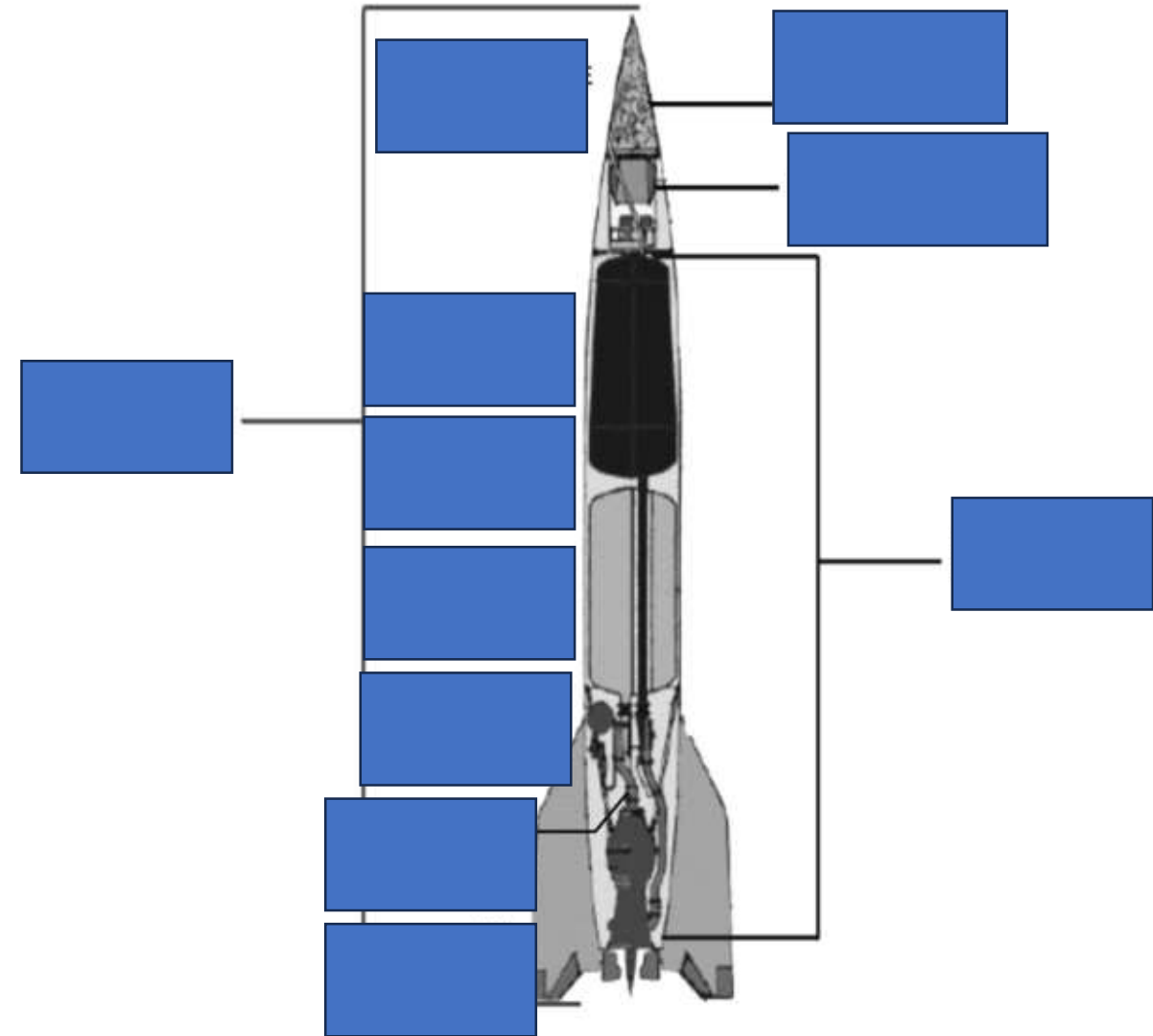
**Nozzle**

**Fins**

- Q2. What are the 4 systems in a rocket?

**structure system, propulsion system, payload system, and guidance system**

# TP2 Name the Parts of a Rocket



EOC240.03

IDENTIFY PARTS OF A  
ROCKET

